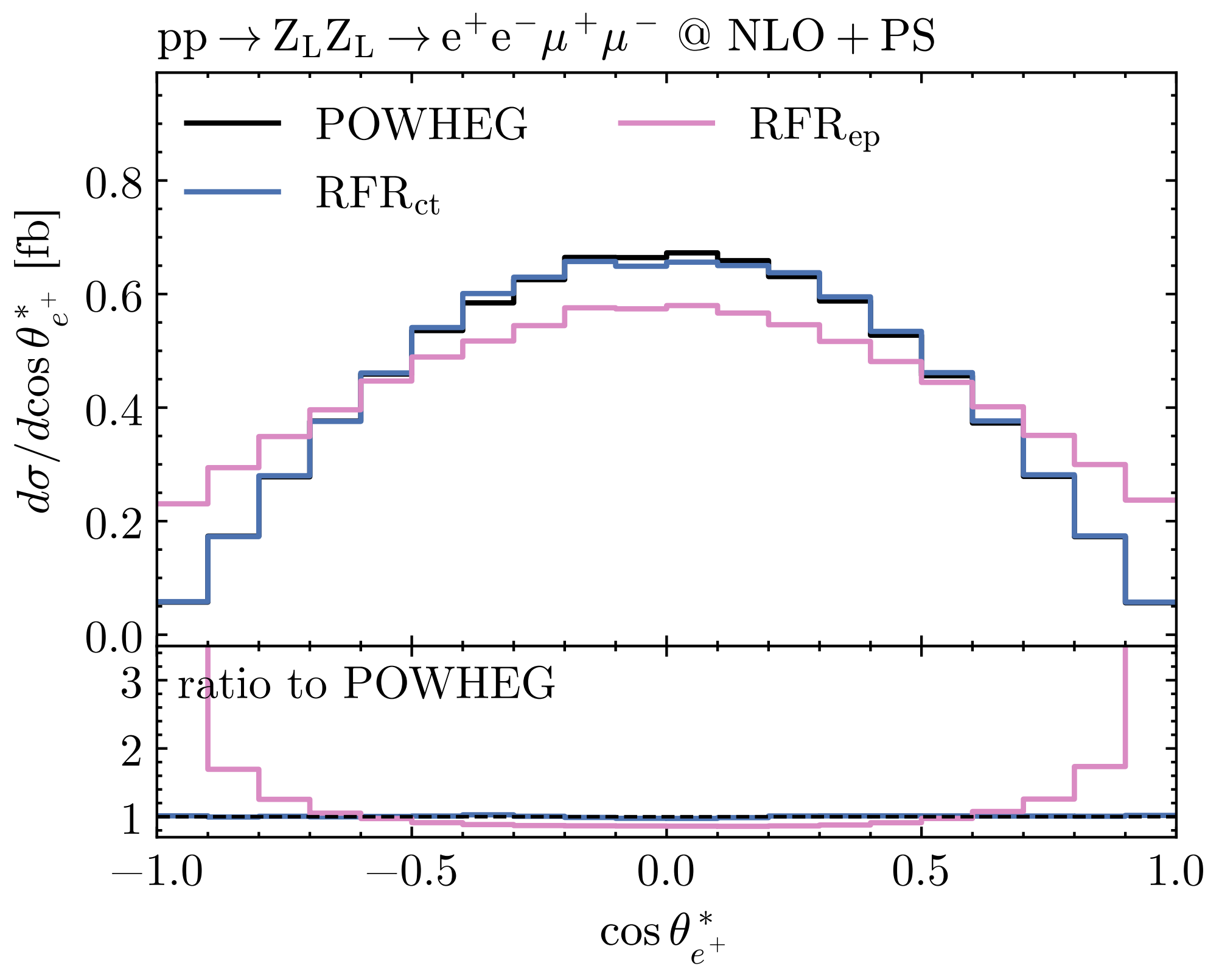
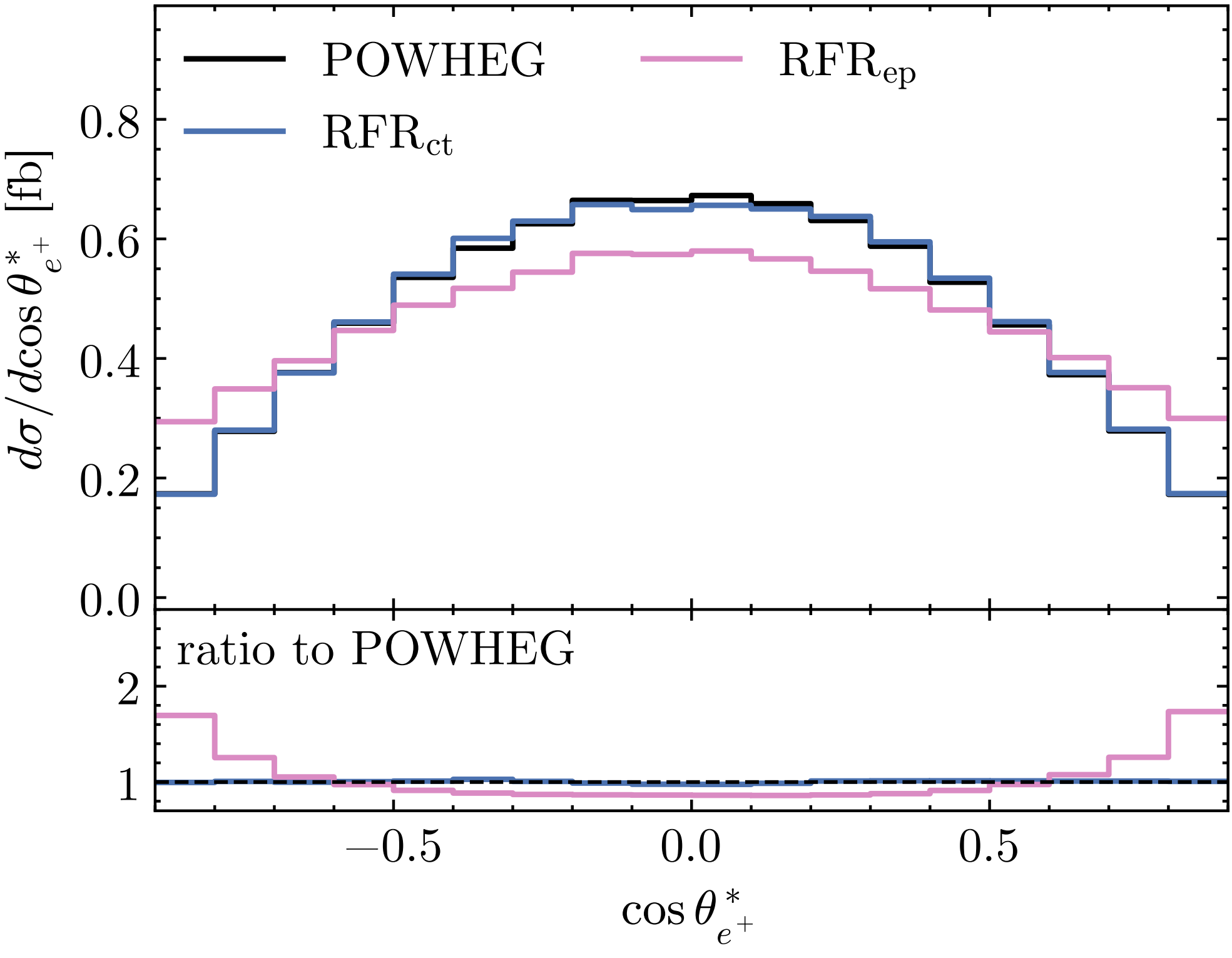
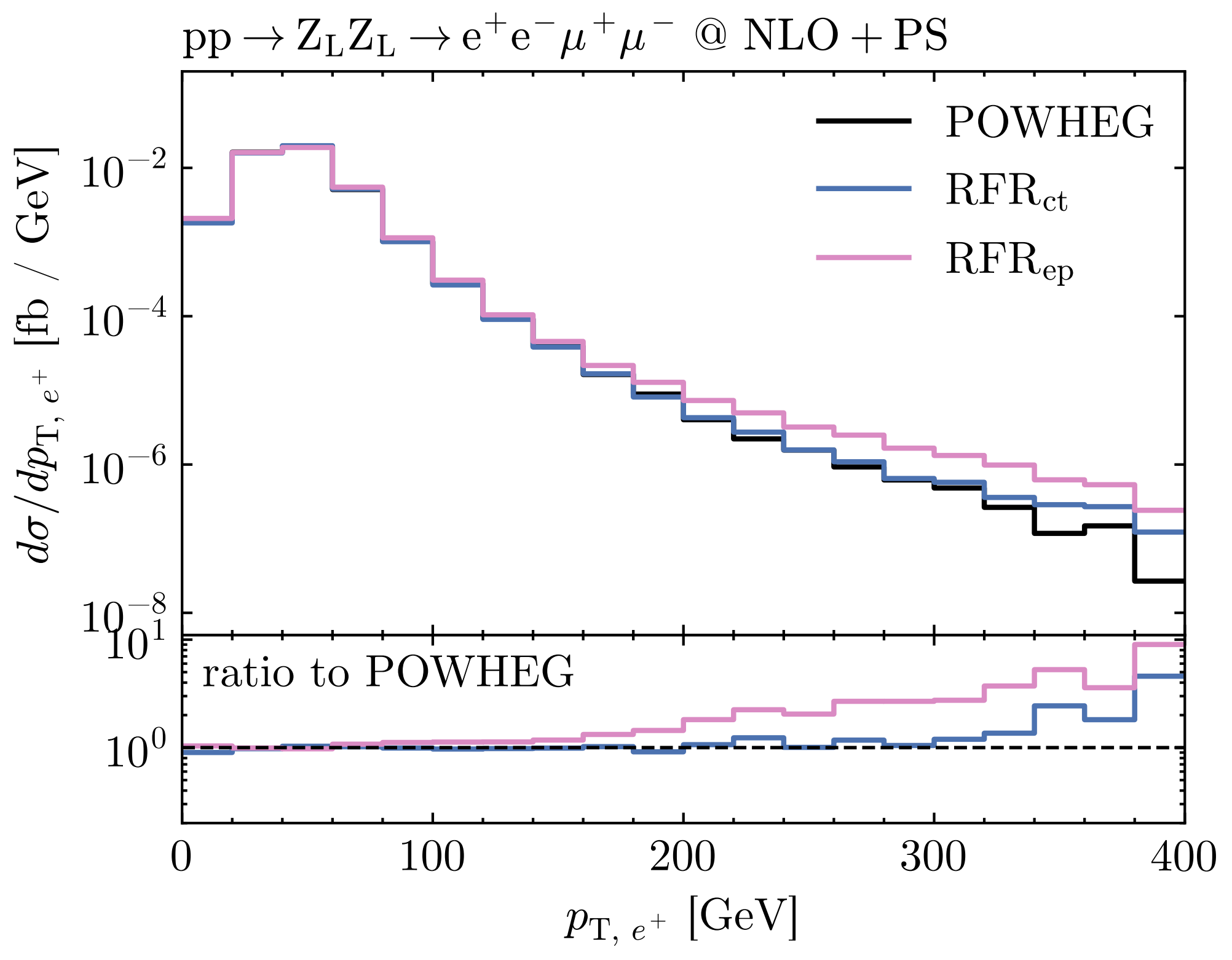
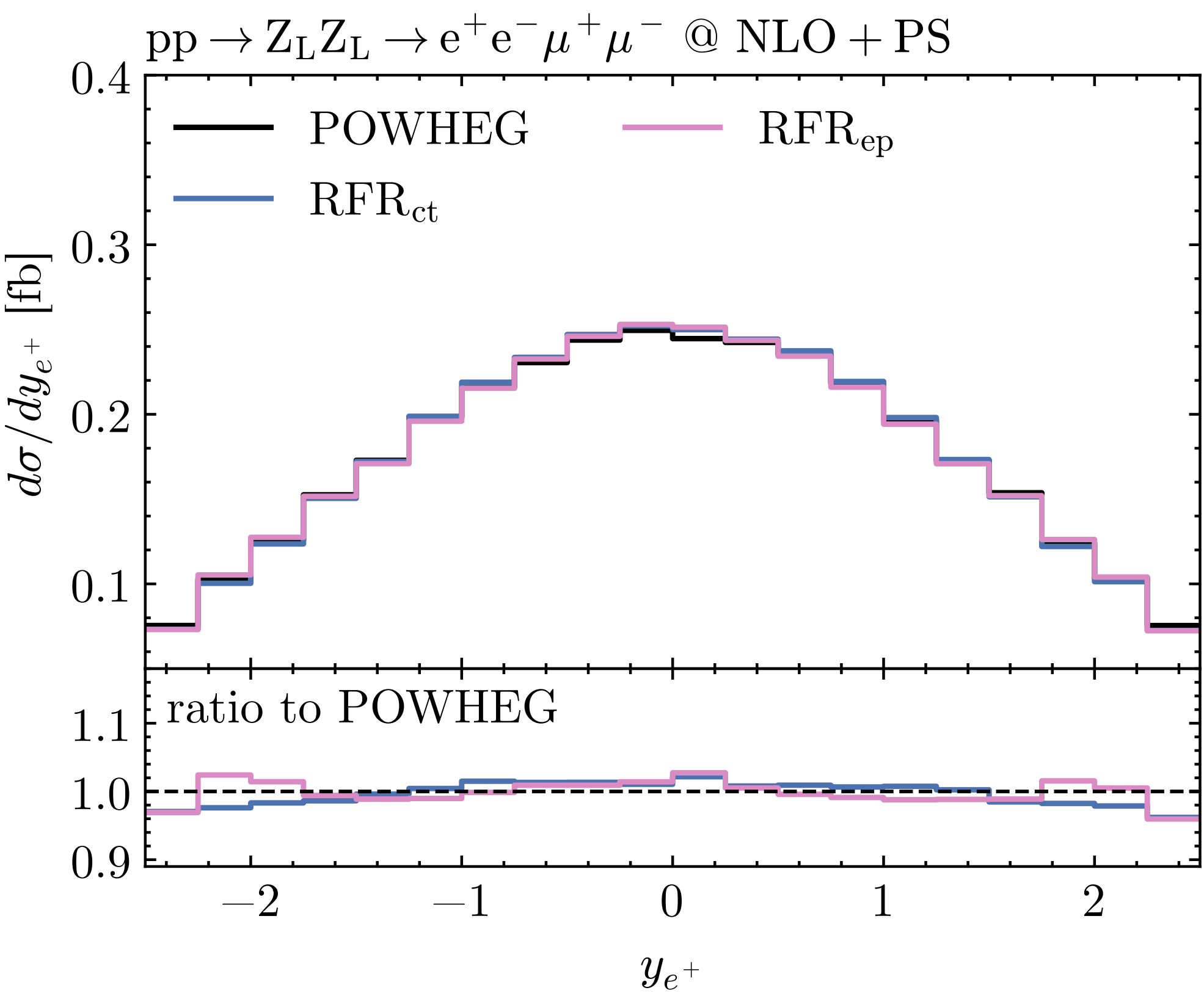
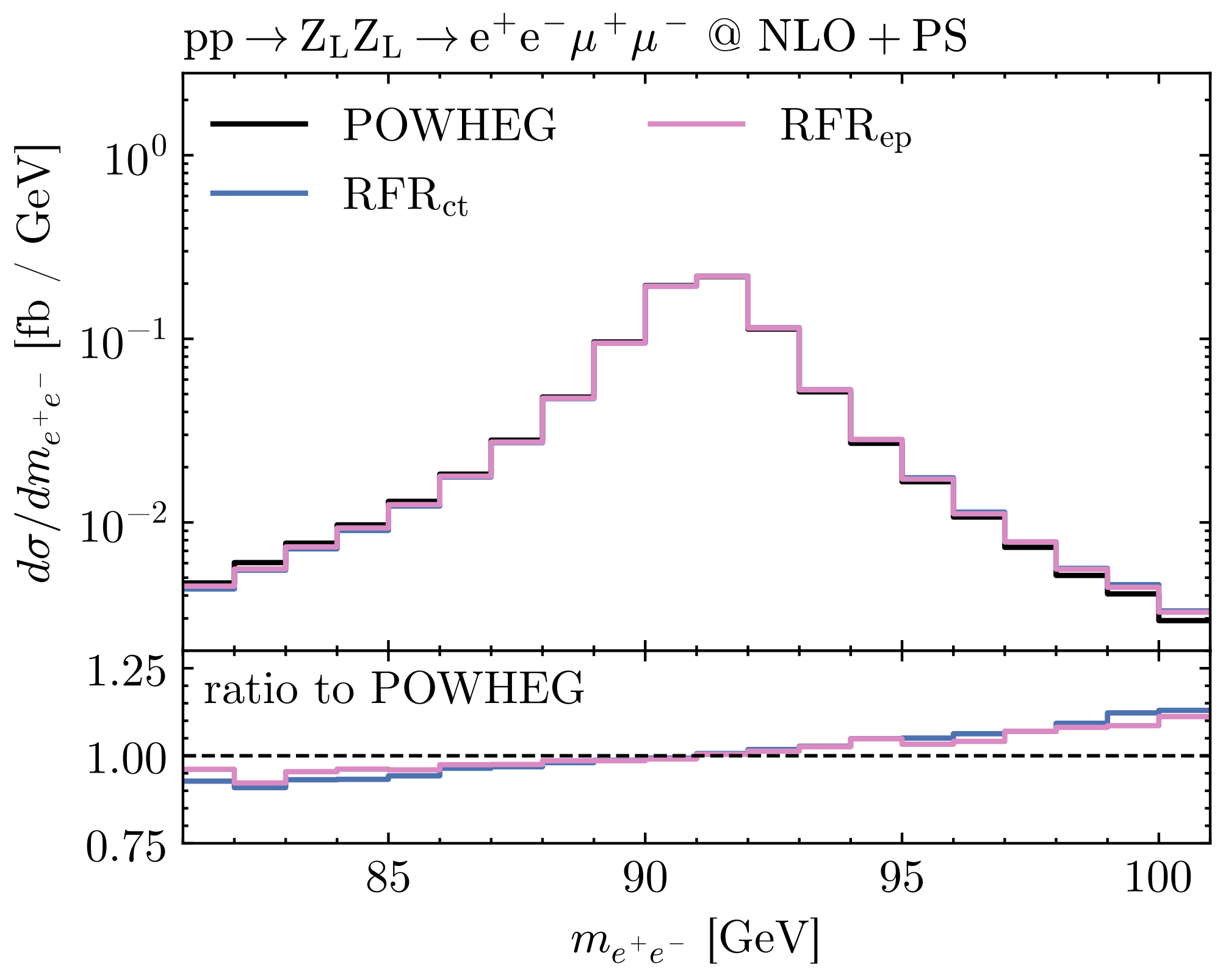
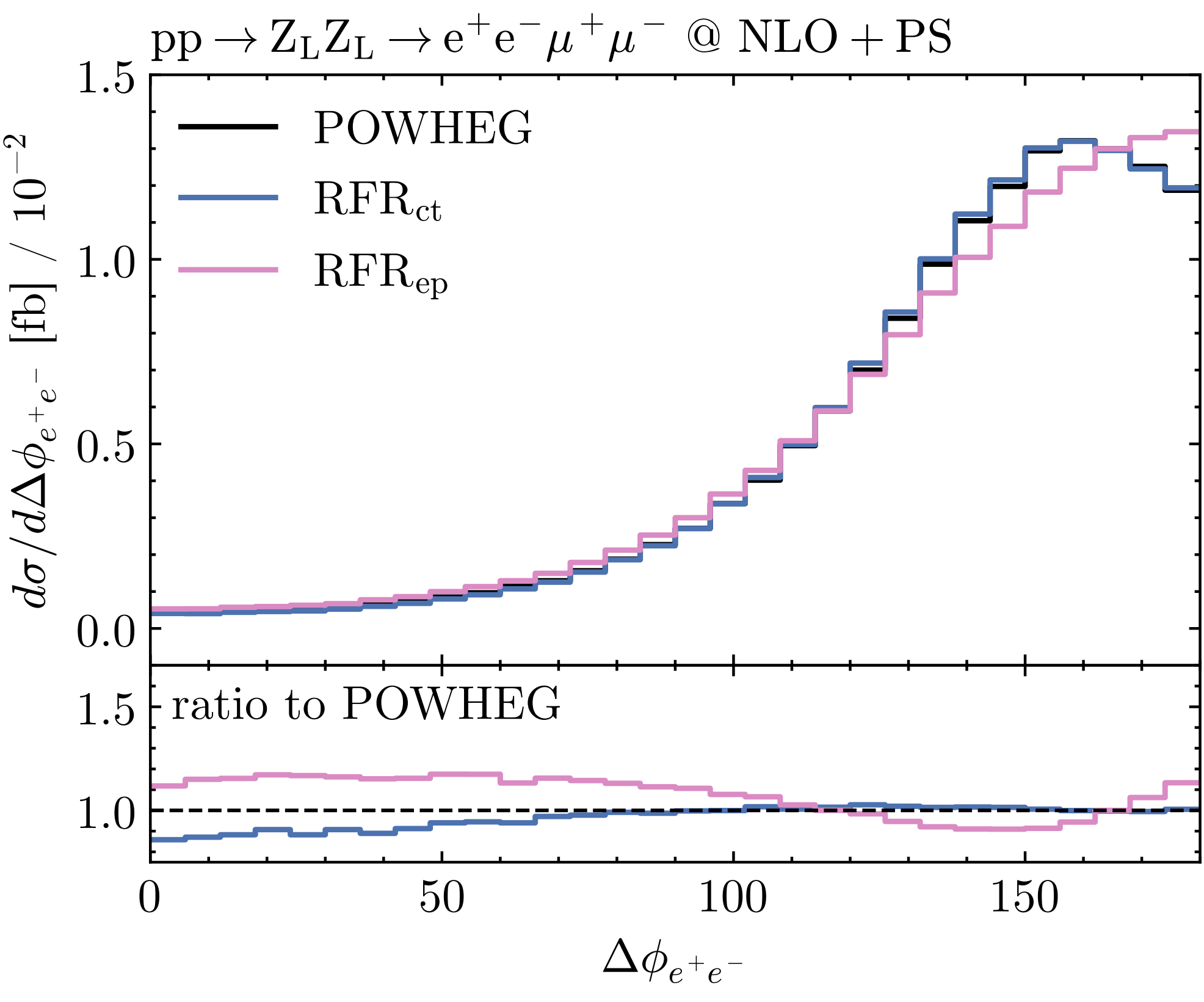

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
POWHEG
$\mathrm{RFR_{ep}}$
$\mathrm{RFR_{ct}}$
$d\sigma/d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to POWHEG
1
2
3
−1.0
−0.5
0.0
0.5
1.0
$\cos\theta^*_{e^+}$

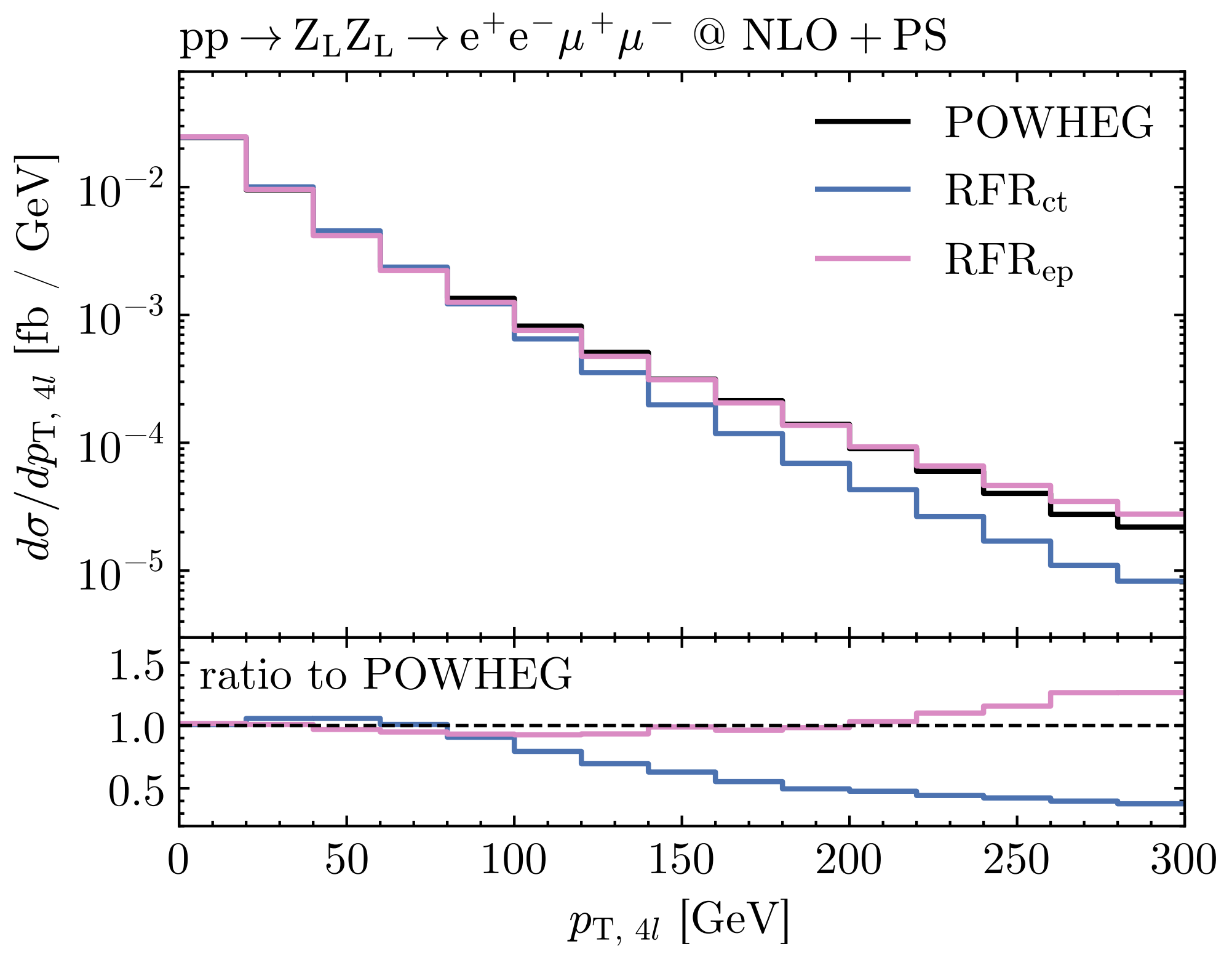

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
POWHEG
RFR$_{\mathrm{ep}}$
RFR$_{\mathrm{ct}}$
$d\sigma/d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to POWHEG
1
2
−0.5
0.0
0.5
$\cos\theta^*_{e^+}$

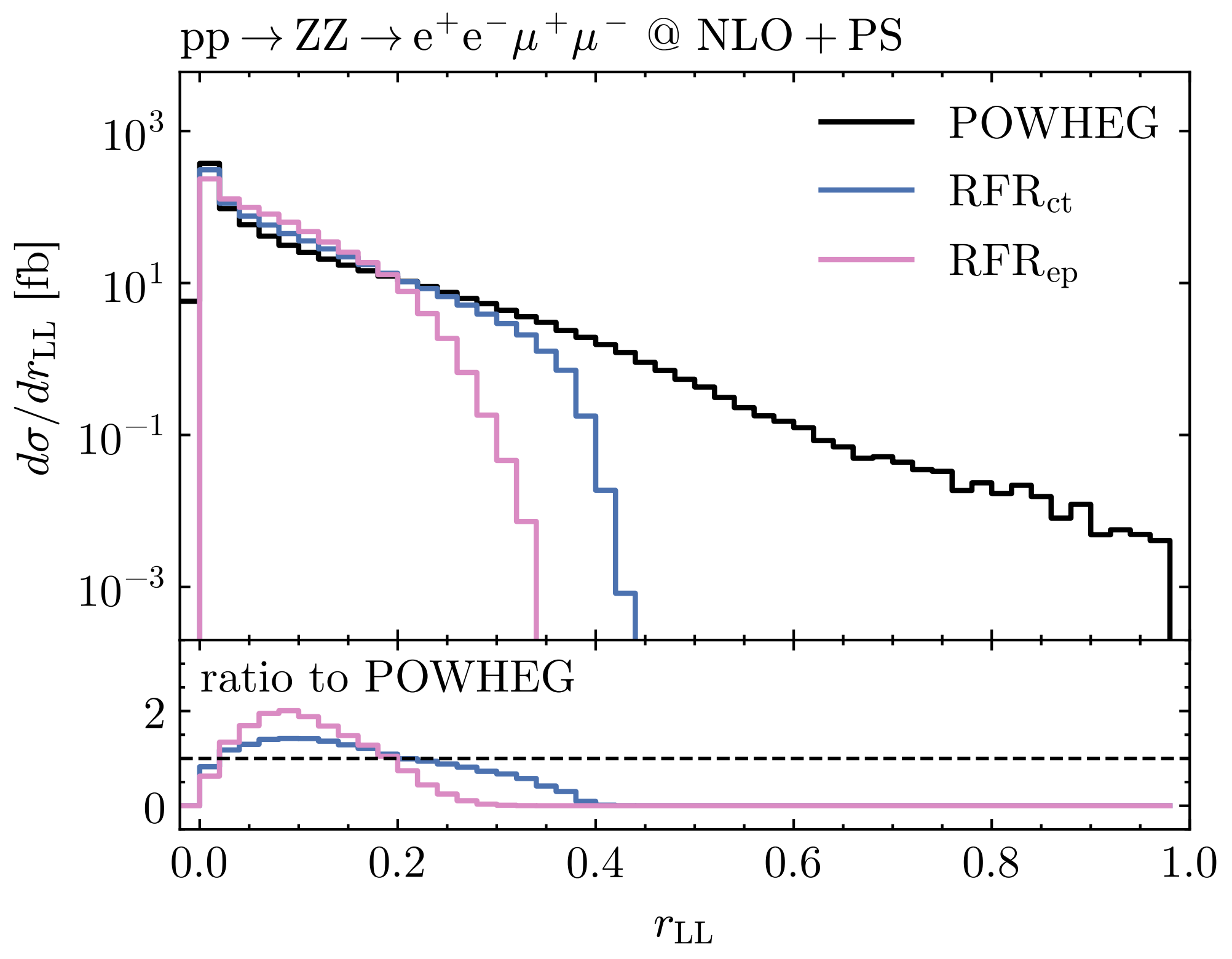

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
POWHEG
$\mathrm{RFR_{ct}}$
$\mathrm{RFR_{ep}}$
$d\sigma/dp_{\mathrm{T},\,e^+}$ [fb / GeV]
10^{-2}
10^{-4}
10^{-6}
10^{-8}
ratio to POWHEG
10^{1}
10^{0}
0
100
200
300
400
$p_{\mathrm{T},\,e^+}$ [GeV]

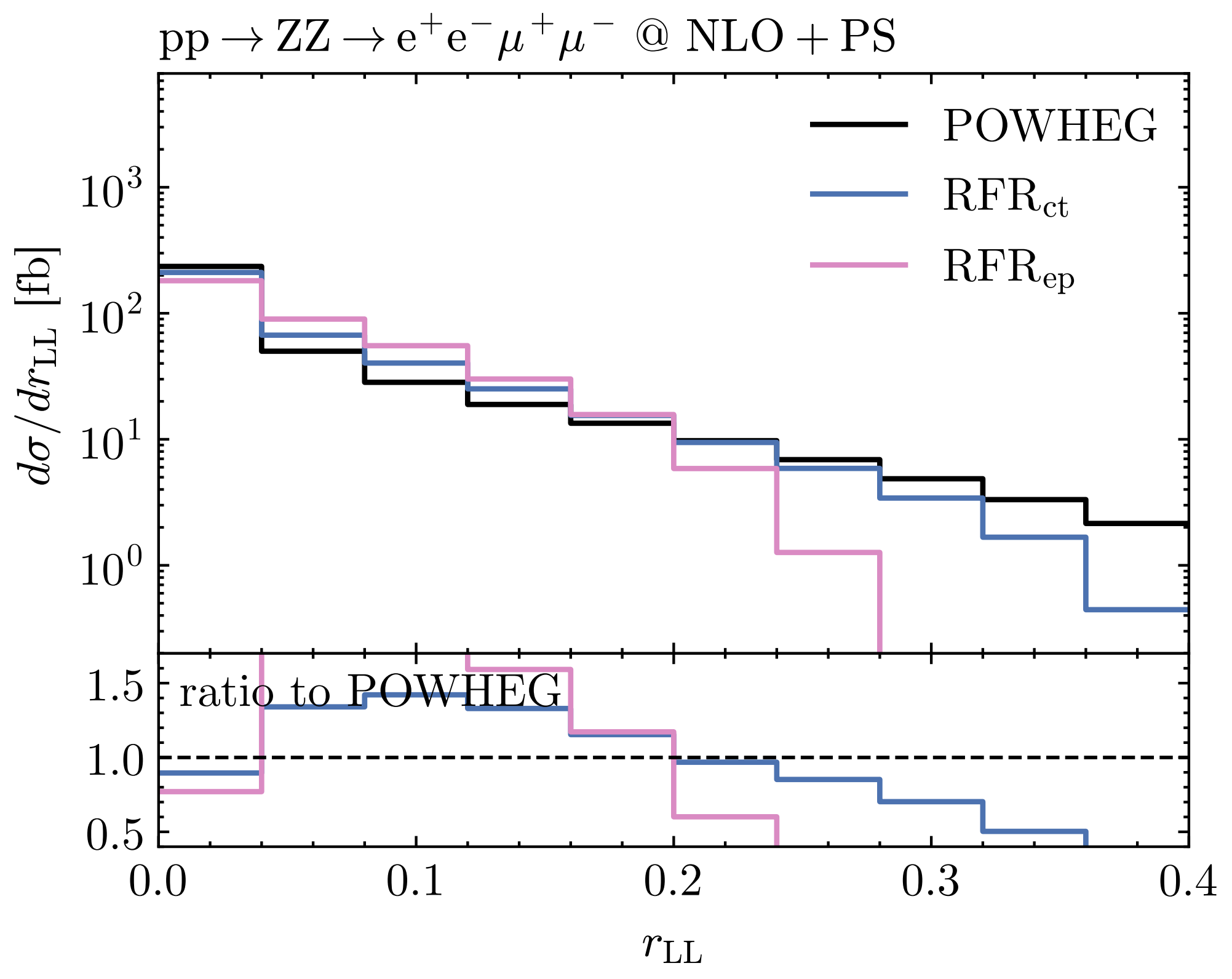

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
POWHEG
$\mathrm{RFR_{ep}}$
$\mathrm{RFR_{ct}}$
$d\sigma/dy_{e^+}$ [fb]
0.4
0.3
0.2
0.1
ratio to POWHEG
1.1
1.0
0.9
−2
−1
0
1
2
y_{e^+}


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
POWHEG
RFR$_{\mathrm{ep}}$
RFR$_{\mathrm{ct}}$
$d\sigma/dm_{e^+e^-}$ [fb / GeV]
10^0
10^{-1}
10^{-2}
ratio to POWHEG
1.25
1.00
0.75
85
90
95
100
$m_{e^+e^-}$ [GeV]


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
POWHEG
$\mathrm{RFR_{ct}}$
$\mathrm{RFR_{ep}}$
$d\sigma / d\Delta\phi_{e^+e^-}$ [fb] / 10^{-2}
1.5
1.0
0.5
0.0
ratio to POWHEG
1.5
1.0
0
50
100
150
$\Delta\phi_{e^+e^-}$


pp → Z_L Z_L → e⁺e⁻μ⁺μ⁻ @ NLO + PS
POWHEG
RFR_ct
RFR_ep
dσ/dp_T, 4l [fb / GeV]
10⁻²
10⁻³
10⁻⁴
10⁻⁵
ratio to POWHEG
1.5
1.0
0.5
0
50
100
150
200
250
300
p_T, 4l [GeV]


pp → ZZ → e+e−μ+μ− @ NLO + PS
POWHEG
RFRct
RFRep
dσ/drLL [fb]
10^3
10^1
10^−1
10^−3
ratio to POWHEG
2
0
0.0
0.2
0.4
0.6
0.8
1.0
rLL


pp → ZZ → e+e−μ+μ− @ NLO + PS
POWHEG
RFRct
RFRep
dσ/drLL [fb]
10^3
10^2
10^1
10^0
ratio to POWHEG
1.5
1.0
0.5
0.0
0.1
0.2
0.3
0.4
rLL